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Xu, Guoqiang; Huang, Xingcan; Shi, Rui; Yang, Yawen; Wu, Pengchen; Zhou, Jingkun; He, Xinxin; Li, Jialin; Zen, Yuyang; Jiao, Yanli; Zhang, Binbin; Li, Jiyu; Zhao, Guangyao; Liu, Yiming; Huang, Ya; Wu, Mengge; Zhang, Qiang; Yang, Zhihui; Yu, Xinge

Published in:
Advanced Functional Materials

Published: 26/02/2024

Document Version:
Post-print, also known as Accepted Author Manuscript, Peer-reviewed or Author Final version

Publication record in CityUHK Scholars:
[Go to record](#)

Published version (DOI):
[10.1002/adfm.202310777](https://doi.org/10.1002/adfm.202310777)

Publication details:
Xu, G., Huang, X., Shi, R., Yang, Y., Wu, P., Zhou, J., He, X., Li, J., Zen, Y., Jiao, Y., Zhang, B., Li, J., Zhao, G., Liu, Y., Huang, Y., Wu, M., Zhang, Q., Yang, Z., & Yu, X. (2024). Triboelectric Nanogenerator Enabled Sweat Extraction and Power Activation for Sweat Monitoring. *Advanced Functional Materials*, 34(9), Article 2310777. <https://doi.org/10.1002/adfm.202310777>

Citing this paper

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This is the accepted version of the following article: Xu, G., Huang, X., Shi, R., Yang, Y., Wu, P., Zhou, J., He, X., Li, J., Zen, Y., Jiao, Y., Zhang, B., Li, J., Zhao, G., Liu, Y., Huang, Y., Wu, M., Zhang, Q., Yang, Z., & Yu, X. (2024). Triboelectric Nanogenerator Enabled Sweat Extraction and Power Activation for Sweat Monitoring. *Advanced Functional Materials*, 34(9), Article 2310777, which has been published in final form at <https://doi.org/10.1002/adfm.202310777>. This article may be used for non-commercial purposes in accordance with Wiley Terms and Conditions for Use of Self-Archived Versions. This article may not be enhanced, enriched or otherwise transformed into a derivative work, without express permission from Wiley or by statutory rights under applicable legislation. Copyright notices must not be removed, obscured or modified. The article must be linked to Wiley's version of record on Wiley Online Library and any embedding, framing or otherwise making available the article or pages thereof by third parties from platforms, services and websites other than Wiley Online Library must be prohibited.

Triboelectric Nanogenerator Enabled Sweat Extraction and Power Activation for Sweat Monitoring

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Keywords: wearable electronics, sweat-activated battery, biosensors, sweat extraction, health monitoring; triboelectric nanogenerator

Abstract

Wearable sweat sensors can detect and monitor various substances in sweat, providing valuable information for healthcare monitoring and clinical diagnostics. Recent advances in flexible electronic technologies have enabled the development of wearable sweat sensors that can measure sweat rate and biochemical substances in real time, although several challenges remain, such as power management and sweat extraction issues. Here, we report a passive sweat extraction strategy as well as a self-powered monitoring system (SEMS) designed for sedentary individuals, i.e., elders. The SEMS system comprises a wearable triboelectric nanogenerator (TENG) for sweat extraction, a sweat-activated battery (SAB) as the integrated power source, carbachol-loaded iontophoresis electrodes for sweat extraction, microfluidics with biosensors for detecting physiological information in sweat, and NFC-based wireless microelectronics for data communication, processing, and collection. By tapping the TENG, sedentary people can passively extract sweat based on the iontophoresis process, and thus for the sensors to detect biological information in sweat. The good flexibility of the SEMS device enables real-time and non-invasive detection of sweat analytes in a wearable format. This system offers a new strategy of sweat collection and analysis for the elder group, and therefore can help to understand human physiology and personalize health monitoring deeply.

1 Introduction

Extensive research works toward the development of wearable bioelectronics are attracting more and more attention in the fields of medical monitoring and clinical diagnostics.¹⁻⁸ Non-invasive monitoring feature of wearable electronics on biosensors is one of the key advantages as which can provides clinical relevant data yet user friendly formats.⁹⁻¹² It is predicted that by 2025, 50% of clinical trials will incorporate wearables and sensors.¹³ Sweat is an important body fluid, that is rich in biological and chemical substances such as pH, inorganic ion concentration and moisture, and thus can reflect nutritional and metabolic conditions.^{5,14-16} Recent research advances have resulted in various wearable sweat sensors that can be used for real-time analysis of sweat biomarkers, including electrolytes, metabolites, and heavy metals.¹⁷⁻²¹ Despite the potential of sweat is an easy-to-get source for both on-demand and in situ analysis, its use is still limited due to the inherent inaccessibility of sedentary humans.

Iontophoresis is a non-invasive, needle-free processor for extracting sweat from skin by providing electric current through the skin to ionize or charge molecules and allow their transportation across the stratum corneum.^{11,22,23} However, the requirement of electrical stimulation typically in mA cm⁻² level also arises the requirement of additional power supply, which may cause a burden for thin, soft, and skin-interfaced electronics. To realize passive sweat collection without additional powering burden, it would be interesting to combine self-powering technologies and iontophoresis. The triboelectric nanogenerator (TENG) seems a promising candidate due to its ability to convert mechanical energy to electrical energy based on the synergistic effect of contact electrification and electrostatic induction.^{24,25} TENG has advantages such as low cost, wide range of material sources, light weight, and high efficiency at low operating frequencies.²⁶ Researchers have also explored the use of TENG-induced electrophoresis for drug delivery, such as in vitro drug transfection,²⁷ transdermal delivery systems,^{22,28} and anti-cancer drug delivery,²⁹ which takes advantage of the TENG's ability to generate high electrical fields and induce the

movement of charged particles for precise and targeted drug delivery.

In this work, we report for the first time ever the use of TENG for self-powered iontophoresis-based sweat collection, as well as the first attempt at fully self-powered sweat extraction, sensing, and wireless data transmission based on such iontophoresis. The TENG-based sweat collection module allows for the extraction of sweat from sedentary individuals and its accumulation in the microfluidic channels without extensive activities. This accumulated sweat can then be used to power the sweat-activated battery (SAB) and analyze the biological information in sweat. Advanced mechanics and materials design not only allow the self-powered sweat collection module to effectively extract and accumulate sweat in the microfluidic channels but also rapidly activate the SAB to provide power for the microcontroller unit (MCU) and biosensors. User studies of such a system on sedentary users prove the accuracy of the sweat sensors and the high feasibility of using this self-powered sweat extraction technology with self-powered sweat sensors for continuous healthcare monitoring of the elderly.

2 Results and Discussion

The schematic illustration of the wearable SEMS for on-demand sedentary users is shown in **Fig. 1a**. This system consists of several components that work together to collect and analyze sweat for health monitoring purposes. The first component is a self-powered sweat extraction/collection module, which includes the wearable TENG connected to two iontophoresis electrodes via a rectifier (**Fig. 1b**). These electrodes are printed on a flexible polyimide substrate, and a hydrogel patch with capsaicin ensures stable sensor-skin contact. When the TENG is tapped, carbachol is delivered to the skin through iontophoresis, promoting localized sweating. The collected sweat is then transported through microchannels to the biosensor for detection. The second component is a self-powered SAB, which is placed on either side of the microchannel structure. When the SAB absorbs sweat, it could generate enough electricity to power the

MCU for data acquisition, processing, and storage of multiplexed monitoring of key metabolic biomarkers in sweat. The third component comprises various sensors positioned on three electrodes attached to the backside of a flexible printed circuit board (FPCB) (**Fig. 1c-I**). It should be noted that one of electrodes serves as a reference electrode. The remarkable flexibility of the FPCB allows for customization according to specific required design (**Fig. 1c-II**). These sensors measure different physiological parameters such as sodium ion (Na^+) concentration, potassium ion (K^+) concentration, and pH values. Finally, a microelectronic component has been developed that is capable of processing the sensor data and facilitating NFC communication with external devices. The device's overall structure is encapsulated in polydimethylsiloxane (PDMS) to ensure a secure fit on the surface of skin (**Fig. 1c-III**).

Fig. 1b shows the schematic diagram of the self-powered sweat extraction process, where a TENG (see **Supporting Information Note 1**) is used to provide electrical field to penetrate skin for sweat activation, which is a self-powered electrophoretic process. To achieve maximum power output of the TENG within limited space, a layer-stacking architecture was designed for the TENG. Gold (Au, 35mm×35mm, thickness of 100 nm) thin film was deposited on a polyimide (PI, 25 μm) film, and a Fluorinated Ethylene Propylene (FEP) film partially covering the electrode serves as the triboelectrification layer (detail can be found in **Supporting Information Note 1**). To evaluate the performance of the TENG under different pressures, the voltage and current outputs of the TENG were measured at pressures ranging from 1 N to 9 N (**Figs. 2a-b**), where it can be found both voltage and current increase with the pressure increase with a maximum power output of 240 V and 17 μA , respectively. Such high-power density (0.23 -0.44 W/m^2) of the TENG can be achieved under light tapping at frequency of 2 Hz, which offers great practical application possibility as 2 Hz tapping can be easily achieved by users (**Fig. 2c**).

The formulation of compounds loaded in the iontophoresis process also plays significant role for sweat extraction. Here the chemicals have been loaded in hydrogels for this purpose, and we conducted

detailed studies the modulation of the sweat rate on the variations of the compounds. In **Supporting Information Note 2**, we utilized the fluorescent small molecule Rhodamine B to demonstrate penetration process on porcine skin, indicating that the application of electrical stimulation from the TENG can enhance the entry of drug molecules into the epidermis.^{23,30,31} Moving forward, in **Supporting Information Note 3**, where a ring-shaped iontophoresis module was used, consisting of carbachol-loaded hydrogel on the anode and NaCl-loaded hydrogel on the cathode, while the TENG provided electrical stimulation. The results show that the sweating volume directly relates with the loaded force, where loading forces from 1 to 9 N includes sweat volume from 2.7 to 5.9 μL of sweat per square centimeter within 30 minutes, as shown in Fig. 2d. Furthermore, different concentrations of carbachol (1%, 3%, 5%) were tested and compared to study their sweat extraction rate differences, with a tapping force of 5 N (Fig.2f). High concentrations of carbachol were found to be more effective in stimulating sweat secretion. It should be noted that although the safe concentration of carbachol is uncertain, the highest concentrations recorded in literature related to iontophoresis are 10% and 12.5%, respectively.^{11,32,33} The dosage we used in the experiment was much lower than these values, and none of the participants showed any discomfort. The sweat extraction rates at different pressures and carbachol concentrations after ten minutes were found to be approximately 1-3 $\mu\text{L}/\text{cm}^2$, as shown in **Fig. 2f**. These findings verified the feasibility of TENG-based stimulation for sweat activation/extraction.

To maximize the efficiency of low-volume sweat analysis and improve the temporal resolution of wearable sensing, a compact and flexible microfluidic module was carefully designed to isolate sweat sampling areas from iontophoresis gels. Numerical simulations were performed to optimize the geometric design of the microfluidic module, including inlet number, orientation, and flow direction with respect to the reservoir geometry (See **Supporting Information Note 4**). With an optimized design for inducing and sampling sweat, the multi-inlet microfluidics system allows for quick and convenient local induction of

sweat, as well as rapid sampling over short periods of time. The microfluidic enables efficient collection of sweat samples for analysis and testing purposes. **Fig. 2g** shows the sweat filling the microchannel at each inlet flow rate of 1.5 $\mu\text{L}/\text{min}$. According to the simulation results, it only takes about 40 seconds to full fill the microchannel. **Fig. 2h** shows the feasibility of collecting sweat by injecting red ink under the test, which are consistent with the theory.

The TENG can effectively generate stimulation to skin for sweat extraction, however, the transient pulse nature from the TENG generated power also brings the hurdles of continuous power supply for electronics. So, a SAB was introduced in the power system where the TENG served as the ignition, and thus the TENG triggered SAB powering module can stably provide power output (**Fig. 3**). Silver oxide (Ag_2O) coated the carbon cloth serves as the cathode of the SAB for catalyzing the oxygen reduction, while the magnesium (Mg) foil with a size of 10 mm $\times$ 10 mm serves as the anode. It was found that the addition of Ag_2O to the cathode significantly increased the current by over 8 times, attributed to its excellent catalytic performance for the oxygen reduction reaction (**Fig. 3a**). Moreover, the distance between the cathode and anode, and the thickness of the Mg anode foil, also can directly influence device size, operation time, and output performance. As shown in **Figs. 3b-c**, short distances between the anode and cathode would increase the short-circuit current of the SAB due to the decreased impedances. At a constant Mg foil thickness of 100 μm and ambient temperature of 25 $^\circ\text{C}$, the working duration of the SAB and the corresponding average short-circuit current drops from 3.52 h and 14.98 mA as the distance between electrodes increases from 0.5 to 2.5 cm, where the result is owing to the increase of the inner impedance of the battery. In order to balance the lifetime and the flexibility of the SAB, 50 μm Mg foil can be found to be the optimized number for the anode. Such SAB could exhibit a maximum current density of 33 mA/cm^2 and maximum power density of 24 mW/cm^2 , and therefore is sufficient to power the electronics (**Fig. 3d**). **Fig. 3e** shows the open-circuit voltage of a representative SAB cell as a function of time after absorbing sweat, where the voltage stabilizes

at 1.66 V- 1.86 V for more than 5 h. Furthermore, the discharge capacity of the SAB reaches 0.5 mA h in such a small cell (1cm×2cm) as its output voltage dropped from 1.74 V to 0 V. To evaluate the effect of absorbed sweat volume on SAB electrical performance, we measured the open-circuit voltage of the batteries by increasing artificial sweat volume gradually from 0 to 6 $\mu\text{L}/\text{cm}^2$ (**Fig. 3g**). The results show that the battery would be easily activated by a very small amount of sweat volume of 2 $\mu\text{L}/\text{cm}^2$, and the corresponding open-circuit voltage of 1.2 V. In addition, The SAB exhibits great softness and stable performance under various mechanical deformations. The soft materials and simple device structures allow the SAB to maintain a stable output voltage of approximately 1.85 V even after being bent and twisted at least 50 times (**Fig. 3h**). In the dynamic mechanical test, the SAB's output voltage slightly dropped from 1.74 V to 1.52 V under 180 ° folding in the middle, while the device could recover to its initial electrical performance after release (**Fig. 3i**). Similarly, the twisting test results also demonstrate the excellent flexibility and robustness of the SAB, as only small and recoverable jitters appeared in the voltage during the twisting process (**Fig. 3i**).

Maintaining proper electrolyte balance plays a crucial role in overall health and well-being. This equilibrium is particularly vital for elderly individuals, as they often face age-related health conditions such as diabetes, kidney disease, and dehydration. Fortunately, by harnessing the power of analyzing ion information in sweat, valuable insights can be gained into these conditions.³⁴⁻³⁶ In **Fig. 4**, we have developed a flexible integrated sensor patch for detecting the concentrations of Na^+ and K^+ as well as the pH value in sweat at room temperature, thereby reflecting the health status of the subjects. The Na^+ sensor consists a layer of poly(3,4-ethylenedioxythiophene:poly(sodium 4-styrenesulfonate) (PEDOT: PSS) on the Au electrodes to decrease the potential drift^{18,37-39}, and a layer of ionophore serving as the sensing area for Na^+ . As shown in **Figs. 4a-b**, the voltage outputs of Na^+ sensor increase with the increase of Na^+ concentration in electrolyte, and there is a linear relationship between the voltage values and the Na^+ concentration. Besides,

the great selectivity of ionophore allow the sensor to exhibit great anti-interference ability (**Fig. 4c**). There are no obvious output responses when adding potassium chloride (KCl) and glucose into the electrolyte, in turn, only the analyte (NaCl) can cause significant change of the sensing signal. **Figs. 4d-e** show the linear relationship between voltage outputs of the polyaniline (PANI) based pH sensor in a large range of 3 to 9. With the decrease of the hydrogen ion (H^+) concentration in electrolyte, the pH value increases and the conductivity of the PANI also decreases, leading to lower voltage outputs.⁴⁰ It's worth to mention the voltage fluctuation caused by interferents is negligible (**Fig. 4f**). Similar with the Na^+ sensor, the K^+ sensor also associates with PEDOT: PSS and corresponding ionophore film. As shown in **Figs. 4g-i**, the outputs of the K^+ sensor are linearly related with the K^+ concentrations in a range of 1 mM to 100 mM with good anti-interference ability and selectivity. All sensors show stable outputs with polyvinyl butyral (PVB)-coated reference electrode that can maintain a stable potential in solutions with different ionic strengths.^{23,41} Besides, long-term stability of biosensors can be obtained for **Supporting Information Note 5**. They were tested on the first, second, and third days following their manufacturing. The results demonstrate that the drift observed over time is comparatively minor when compared to the variations induced by changes in concentration. This finding highlights the high stability exhibited by our biosensors. Overall, these sensors on the integrated flexible patch all exhibit excellent linearity, selectivity, and sensitivity, which would largely increase the detection accuracy of human sweat.

The wearable TENG, as shown in **Fig. 5a**, is a versatile device that can take various forms such as textile⁴², insole²⁴, wristband⁴³, hand⁴⁴, and arm pressing⁴⁵, can convert biomechanical energy into electrical energy, making it an ideal solution for self-powered wearable systems. It is particularly useful for individuals who cannot sweat through intense exercises, such as elderly people or those with physical disabilities. As demonstrated in **Fig. 5b**, tapping the TENG device on their arm allows them to collect TENG energy through localized movements, which can accelerate sweating and enable them to obtain sweat information more

quickly. **Fig. 5c** shows a real-life example of a TENG device placed on the arm of a subject for self-powered sweat collection and battery activation, and the sensors can be powered by the battery and the data can be wirelessly transmitted by NFC. The data collected by the sensor patch is converted into digital signals that are stored in a memory via I2C, and can be transmitted to a mobile terminal through the NFC chip. The GUI in the mobile phone can display and store the received data for further analysis. It should be emphasized that here we only use layered TENG⁴⁶, has a very small volume and high current output, as a power source to demonstrate the effectiveness of our device. In practical applications, TENG can take on various forms to provide energy for electrically stimulated sweating. **Fig. 5d** illustrates the impact of the TENG stimulation on the pH, Na⁺ concentration and K⁺ concentration of sweat. The results demonstrate that TENG stimulation enables the detection of sweat information in approximately 8 minutes with a 3% concentration of carbachol. Subsequently, we proceeded to monitor continuously for approximately 20 minutes. In the control group, where no electrical stimuli were used under the same conditions, data collection could also commence after about 36 minutes of rest. This suggests that TENG electrical stimulation significantly accelerates the response time for sweat detection and our device possesses a certain level of continuous health monitoring capability. Besides, it should be noted that theoretical monitoring time of this system is primarily determined by the energy of the SAB, which powers the microelectronic chip. Throughout this monitoring period, real-time sweat information can be obtained using the mobile app. **Figs. 5e-g** show the consistency of the results collected by the biosensor in different body locations. For different subjects could be obtained from **Supporting information Note 6**. Overall, the self-sweat collection and self-powered device offers a convenient and effective solution for continuous healthcare monitoring in wearable systems.

3 Conclusion

In this paper, we designed a SEMS based on TENG and SAB, which can be used for sedentary people to conquer the difficulty-to-sweat challenges in these groups. By incorporating the concept of TENG-induced stimulation, we were able to accelerate sweating, thereby reducing the original waiting time of 36 minutes to about 8 minutes. In addition, the skin-interfaced SAB with high output power density can be also triggered by the TENG induced sweat, and then powered electronics and biosensors. Furthermore, the pH value, Na^+ concentration, and K^+ concentration in sweat can be detected by our-self developed biosensor and the data can be transmitted to the terminal device through the NFC function. The results indicate that this technology can be used for self-powered sweat collection, sensing analysis and data transmission, and therefore for continuous intelligent healthcare monitoring. It is foreseeable that it will have broad application prospects in the fields of wearable electronics and health monitoring.

4 Methods

Fabrication of TENG: The manufacturing process of the layered TENG is achieved through a symmetrical structure, as shown in **Fig. 1a**. A 100 μm thick polyimide film is folded into 6 layers to serve as the substrate, with each layer measuring 40 mm $\times$ 40 mm. Copper electrodes are plated on both sides of the polyimide film using flexible printed circuit board technology, with an area of 35 mm $\times$ 35 mm. FEP films are alternatively attached to one of the copper electrodes on the same layer, forming a pair of electrodes. The FEP films have a thickness of 25 μm . The other bare copper electrode serves as the output electrode, which together with the FEP-coated electrode constitute the output end of the TENG.

Fabrication of hydrogel patch: The anode hydrogel was prepared by combining 1% carbachol and 3% agarose, based on weight, in deionized water. The mixture was heated to 150°C and stirred until homogenous. Any lost deionized water during the heating and stirring process was replaced, and the solution was further heated to 80°C to ensure even distribution of the added water. The resulting mixture was then poured into a

pre-prepared mold and allowed to solidify. Similarly, the cathode was prepared using the same method, but instead of carbachol, 1% NaCl (w/w) was used.^{10,23}

Fabrication of SAB: The production process of SAB can be described as follows: Firstly, carbon cloth and Mg sheet were cut into a specified size of 1 cm × 1 cm. The next step was to submerge the carbon cloth in an aqueous solution containing graphene and Ag₂O with a weight ratio of 1:1 was uniformly dispersed (**Supporting information of Note 7**), followed by drying at a temperature of 100 °C for 10 minutes. It is important to note that the weight ratio of graphene and Ag₂O was based on the optimal results obtained from our previous related research.¹⁹ The coated carbon cloth was then dried at a temperature of 100 °C for 10 minutes, resulting in highly efficient anode current collectors. It is important to note that the surface of the carbon cloth was subjected to plasma treatment before the coating process, and the Ag₂O/graphene mixture underwent approximately twenty minutes of ultrasonic treatment to ensure complete dispersion. Afterwards, the Ag₂O/graphene-coated carbon cloth and the polished Mg sheet were placed adjacent to each other on the absorbent paper. Finally, a thin layer of cotton containing KCl powders was attached to the electrodes to create the SABs.

Fabrication of multiplexed biosensor Patch: The fabrication of sensor patch started on the FPCB electrodes. For Na⁺ and K⁺ sensors, a layer of PEDOT: PSS was firstly electrodeposited on the gold electrode under a constant current of 141.36 μA for 1200 s. The electrolyte of electrodeposition was the mixed solution of 0.01 M 3,4-Ethylenedioxythiophene (EDOT) and 0.1 M polystyrene sulfonate (NaPSS). Following, 2 μL corresponding ionophore cocktail was dropped on the electrode and stored in refrigerator overnight. The Na⁺ ionophore cocktail included 1 mg Na ionophore X, 0.55 mg sodium tetrakis [3,5-bis(trifluoromethyl)phenyl] borate (Na-TFPB), 33mg polyvinyl chloride (PVC, K-value 72–1), and 65.45 mg bis(2-ethylehexyl) sebacate (DOS) dissolved in 610 μL of tetrahydrofuran. The difference between Na⁺ ionophore cocktail and K⁺ ionophore cocktail was using valinomycin to replace Na ionophore X as the

ionophore. For the pH sensor, the layer of PANI was obtained on gold electrode by electrodeposition using cyclic voltammetry methods for 40 segments (-0.2 to 1 V at a scan rate of 0.1 V s^{-1}). The electrolyte included 0.1 M aniline and 1 M sulfuric acid. Besides, silver (Ag)/silver chloride (AgCl) ink was screen-printed on gold electrode to be the references. $2 \text{ }\mu\text{L}$ of polyvinyl butyral (PVB) cocktail (79 mg PVB, 50 mg NaCl and 2 mg block polymer PEO-PPO-PEO dissolved in 1 mL methanol) was lastly dropped on the Ag/AgCl electrode to minimize the potential drift.^{47,48}

Statistical Analysis: The illustration models were constructed on a Cinema 4D (Maxon Computer). Voltage and Current of TENG were obtained by Keithley 6514 Electrometer (Keithley Instruments, Cleveland, OH). Data of SAB and biosensors were mainly obtained by the data acquisition (DAQ)/multimeter system (Power Lab 16/35, AD Instruments) and electrochemical work station of CHI 660E. Data processing was operated by the Origin software (Origin Lab).

Ethical information for studies involving human subjects

All experiments involving human subjects were approved by Research Committee of City University of Hong Kong and conducted in compliance with the guidelines. All participants for the studies were fully voluntary and submitted the written informed consents to take part in the study. The device is attached on the subjects' body and thus the necessary but limited identifiable images must be used. All identifiable information was totally consented by the user.

Acknowledgements

All experiments involving human subjects were approved by City University of Hong Kong, and conducted in compliance with the guidelines. All participants for the studies were fully voluntary and submitted their informed consent. This work is supported by the Foundation of National Natural Science

Foundation of China (NSFC) (Grant No. 61421002), City University of Hong Kong (Grant Nos. 9678274, 9667221 and 9680322), Research Grants Council of Hong Kong Special Administrative Region (Grant Nos. 21210820, 11213721 and 11215722).

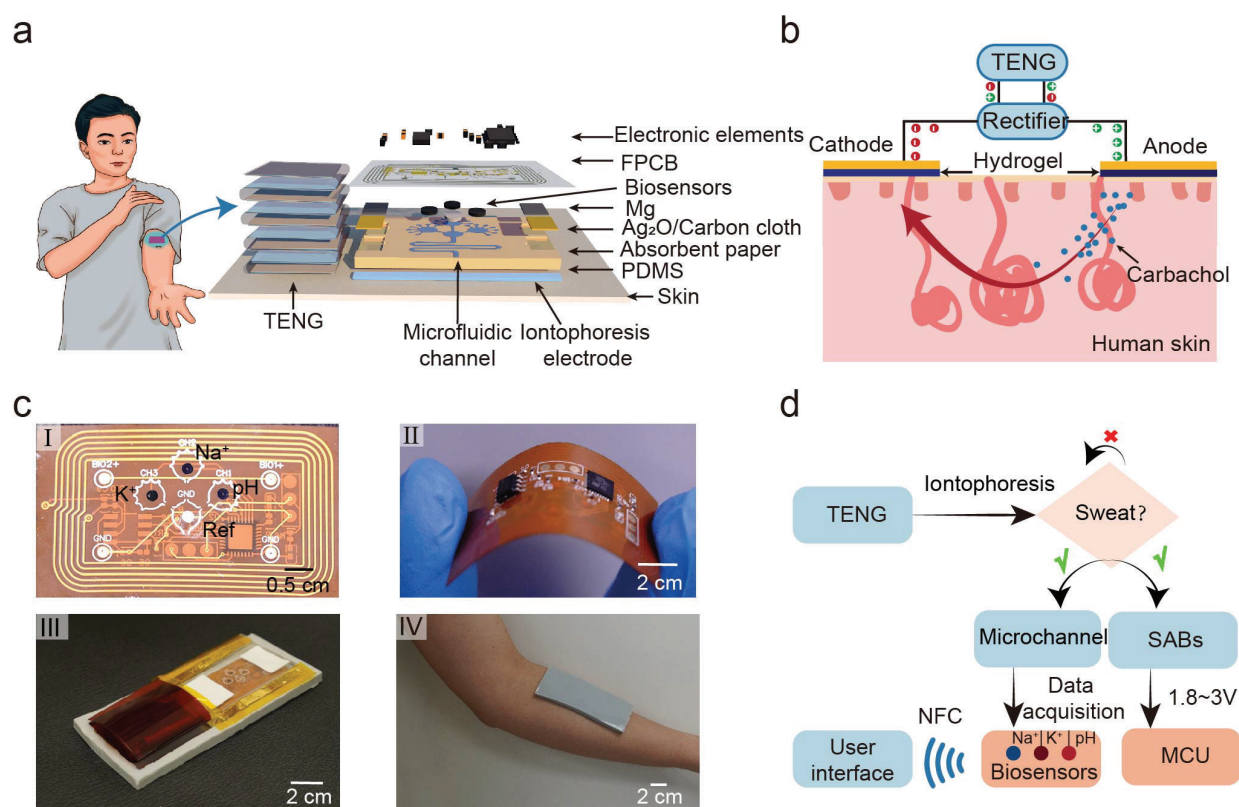


Figure 1. Schematics and images of SEMS. **(a)** Schematic of Wearable SEMS for On-Demand Sedentary Humans. **(b)** TENG-based approach to stimulate sweating. **(c)** Optical images of SEMS Devices. I. Physical Images of Microcircuit with Biosensors. II. Remarkable bending ability of FPCB. III. Encapsulated TENG and microcircuit in PDMS. IV. Secure fit of device on skin surface. **(d)** Operation Flowchart of the SEMS Device

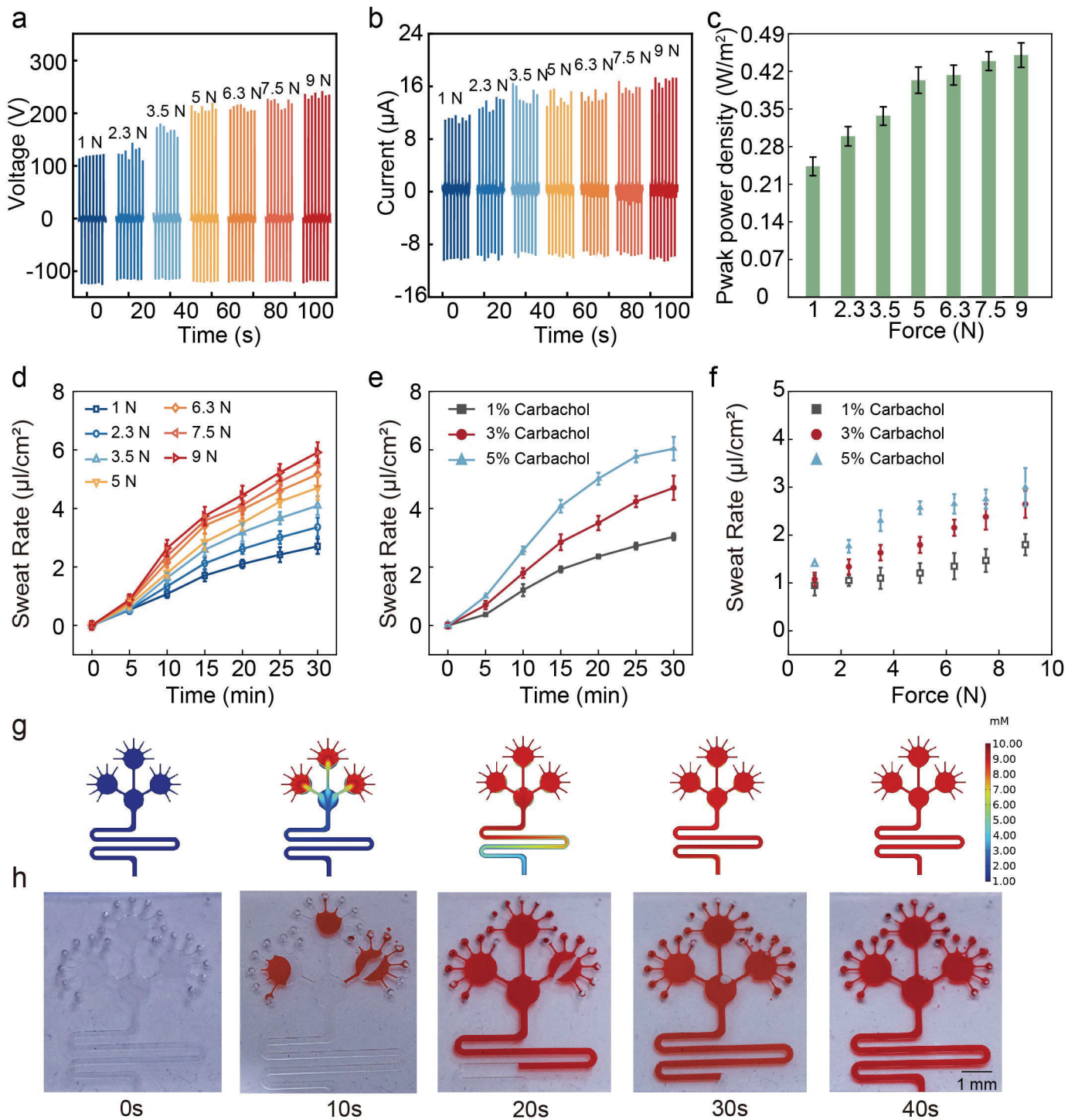


Figure 2. Study of electrophoretic flow driven by TENG. **(a)** TENG voltage output at different pressures. **(b)** TENG current output at different pressures. **(c)** Peak power density of TENG under different pressures. **(d)** Sweat rate under different forces. **(e)** Sweat rate under different carbachol concentration when TENG work with a frequency of 2Hz. **(f)** Sweat extraction rate under different pressures and carbachol concentrations. **(g)** Numerically simulation of the concentration distributions in microfluidic channel at different time instances (inlet flow rate, 1.5 μ L/min). **(h)** Photograph of microfluidic channel by injecting red marked liquid under the test.

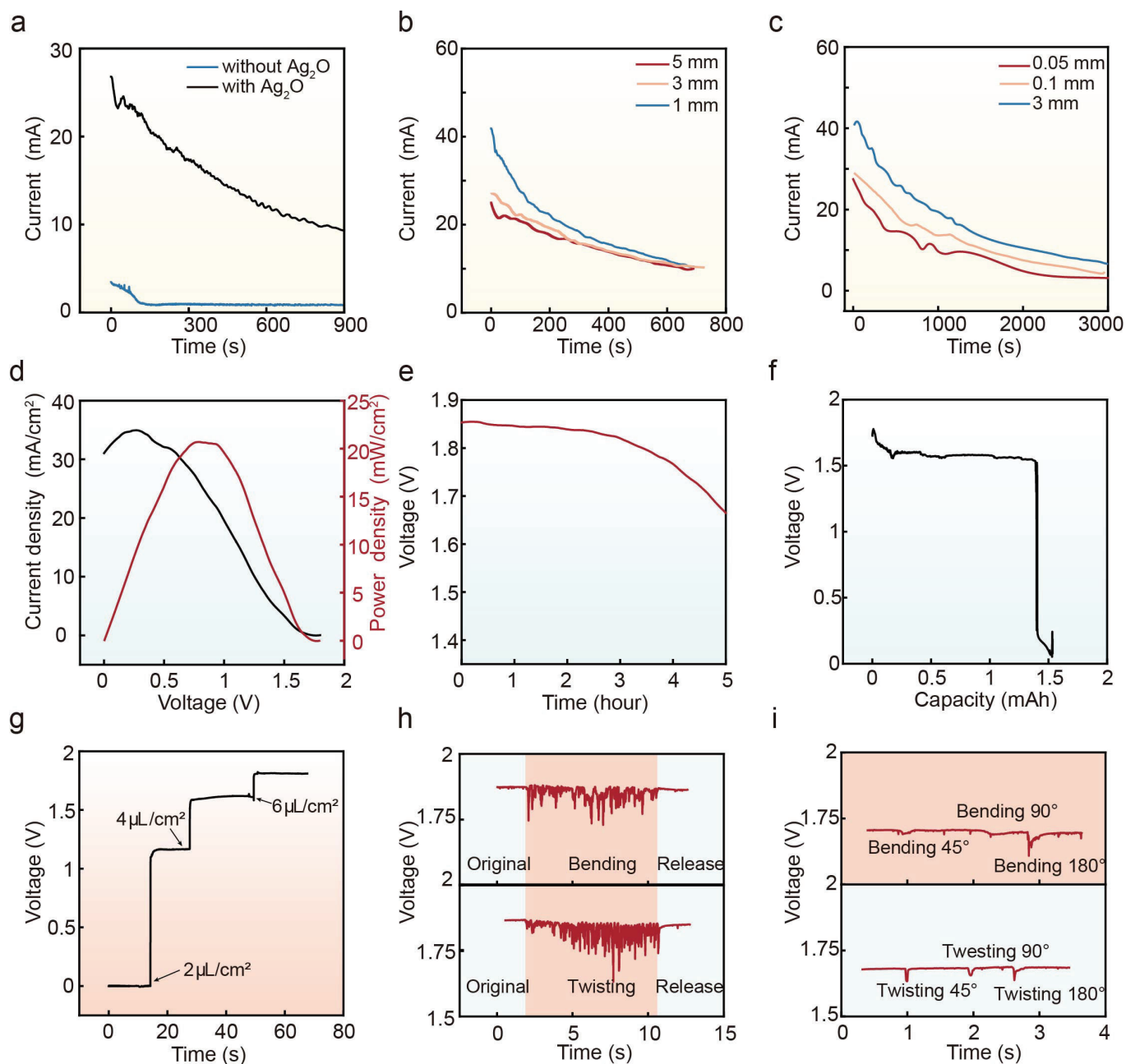


Figure 3. Performance characterization of the SABs. **(a)** comparison of SAB when the anode with or without Ag_2O . **(b)** Short-circuit current of SAB with different electrodes distance. **(c)** Short-circuit current of the SAB with three different thickness of Mg foil. **(d)** Current density and power density curve of the SAB. **(e)** Open-circuit of SAB as a function of time. **(f)** Capacity of SAB. **(g)** Electrical response of the SABs as a function of time with increasing injected sweat volume. **(h)** Electrical output response of the SABs under continuous bending and twisting. **(i)** Electrical output response of the SABs bended as 45° , 90° , and 180° as a function of time.

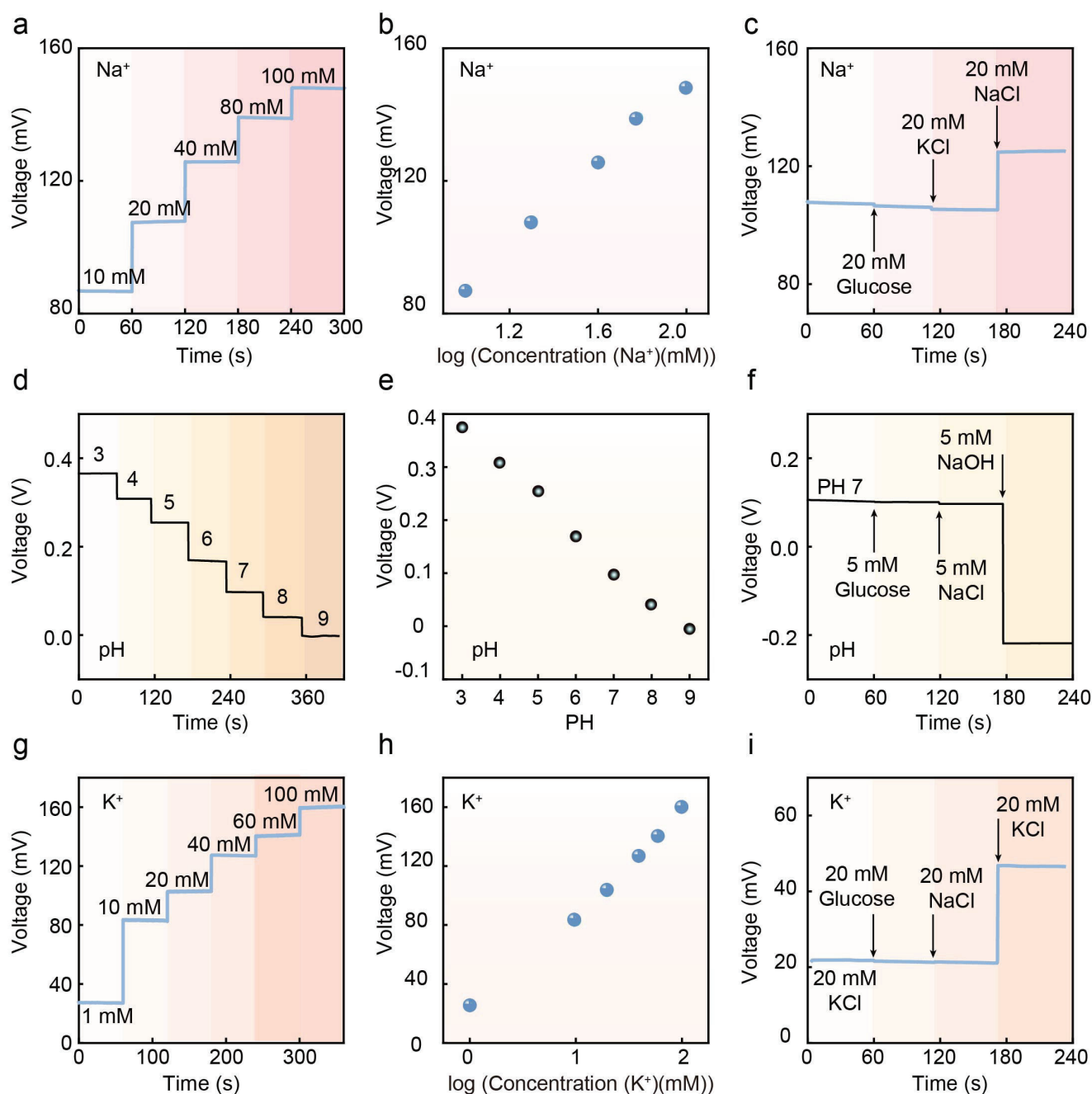


Figure 4. Multiplexed biosensor patch characterization. **(a)** and **(b)** Time-Voltage curve and the corresponding calibration curve of the Na⁺ sensor. **(c)** Anti-interference characterization of the Na⁺ sensor. **(d)** and **(f)** Time-Voltage curve and the corresponding calibration curve of the pH sensor. **(f)** Anti-interference characterization of the pH sensor. **(g)** and **(h)** Time-Voltage curve and the corresponding calibration curve of the K⁺ sensor. **(c)** Anti-interference characterization of the K⁺ sensor.

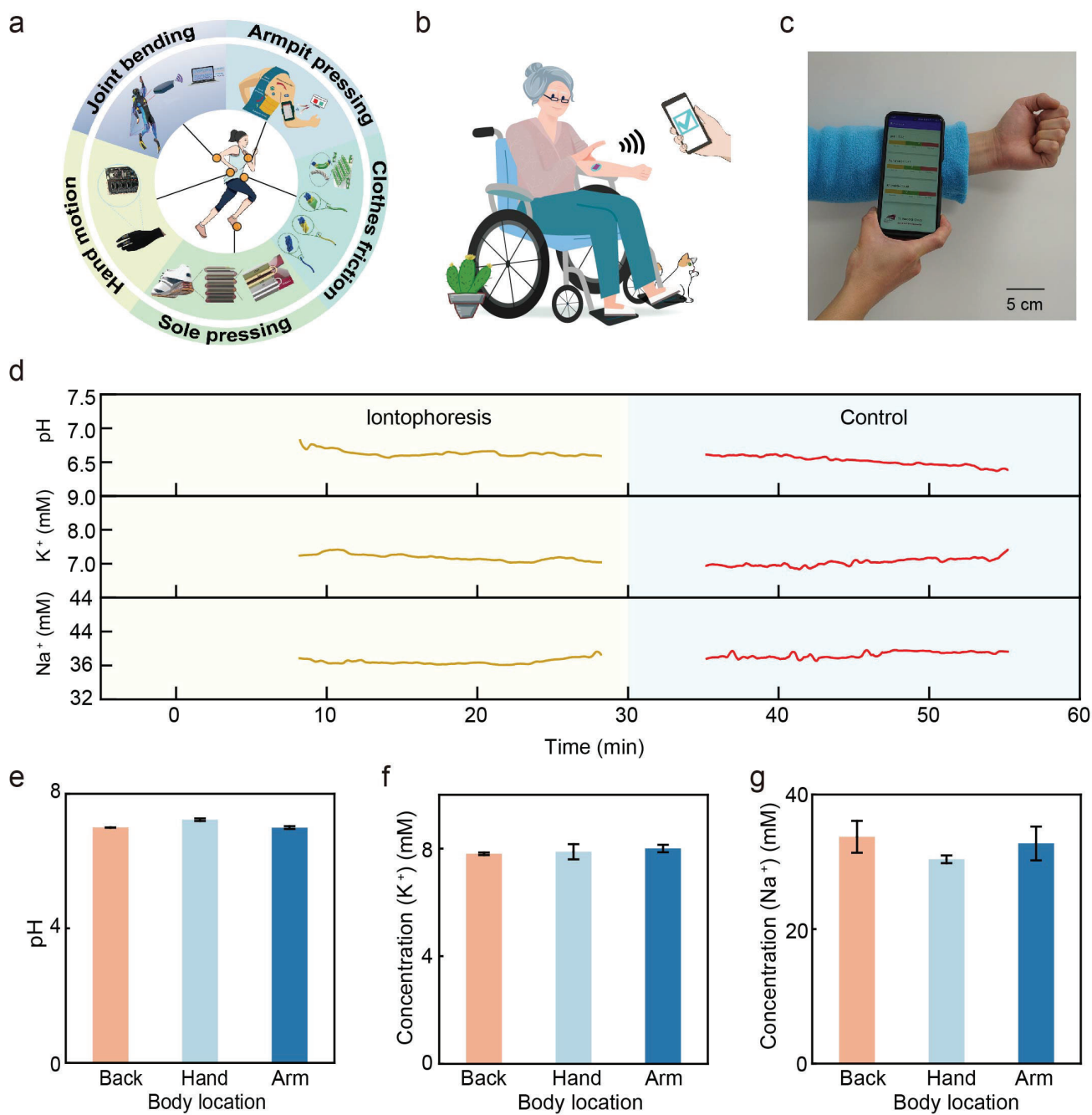


Figure 5. On-body health monitoring and wireless data transmission. **(a)** Versatility of wearable TENG devices and energy collection from different parts of the human body. **(b)** Collection of sweat information through localized movement with TENG devices. **(c)** Obtaining sweat information through NFC functionality with TENG devices. **(d)** Effects of TENG stimulation on pH value, Na^+ concentration, and K^+ concentration. **(e)** pH Value, Na^+ concentration, and K^+ concentration without TENG stimulation. **(f-h)** Consistency of sweat information from different body locations.

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